

# Performance Analysis of Centralized RAN Deployment with Non-ideal Fronthaul in LTE-Advanced Networks

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**Abstract**—Centralized RAN and baseband pooling architectures are increasingly being seen as the way cellular technologies will be deployed in the future due to a number of advantages they provide to the operator. However, LTE and LTE-Advanced were designed with a distributed architecture in mind. Therefore, there are certain inherent technical issues when the link connecting the baseband and RRU (Remote Radio Unit) has a non-negligible latency. In this paper, we study the impact of this non-ideal fronthaul on the downlink and uplink performance of LTE-Advanced networks. We show that, while the impact on the peak throughput is quite severe, it is more modest when we have a larger number of active UEs. The degradation in performance is larger in the uplink than in the downlink due to the synchronous HARQ mechanism used in LTE uplink. In the downlink, we compare the performance of centralized RAN with that of distributed RAN in case of non-ideal fronthaul (FH) and backhaul (BH), respectively, and determine whether it is better to incur backhaul or fronthaul latency from a performance perspective. In the uplink, we analyze the loss in performance due to fronthaul latency. We have developed a novel *look-ahead scheduling* scheme that results in significant improvement in performance, especially when the number of active UEs is small.

## I. INTRODUCTION

When LTE was standardized, a distributed architecture was assumed where the baseband processing happens at the site location as illustrated in Figure 1. In this scenario, the scheduler runs at every eNodeB. However, a number of recent technical developments have triggered an interest in operators to consider a centralized baseband pooling architecture or even a cloud RAN deployment, where the baseband processing for a number of cells happens at a central location that could be several kilometres away from the base-site location as shown in Figure 2. At the base-site location, there is only the RRU (Remote Radio Unit). The advantages of this architecture including the recent advantages in “cloud” technologies have been treated in a number of recent papers [1], [2], [3], [4]. Such an architecture implies that the link between baseband and RRU (also known as fronthaul) should be connected with “dark fiber” with zero latency. However, not all operators can have such fiber links between the baseband and RRU, but would still like to benefit from the advantages of centralized RAN. In Section II of the paper, we study the performance impact of this non-ideal fronthaul on the downlink as a function of fronthaul latency and number of active UEs. We next study the gains from coordinated scheduling [9], [10], which exploits the low-latency interconnects between the

schedulers of different cells deployed in a centralized fashion. Section III compares the loss in performance for both centralized and distributed architectures in the presence of a non-ideal connection between BBU (baseband unit) and RRU for centralized architecture as compared to a non-ideal connection between different neighbors for the distributed architecture. Section IV further studies the performance loss in the uplink because of fronthaul latency. The impact of non-ideal fronthaul is more severe on the uplink because of synchronous retransmissions [5]. Since, synchronous re-transmissions can happen only on predefined TTIs, any fronthaul latency will result in a minimum delay of 8 ms in re-transmissions. To reduce the impact of fronthaul latency in the uplink, we have developed a novel scheduling scheme that improves performance by reducing the number of scheduled UEs per TTI such that all resources can be maximally occupied. Section V concludes the paper.

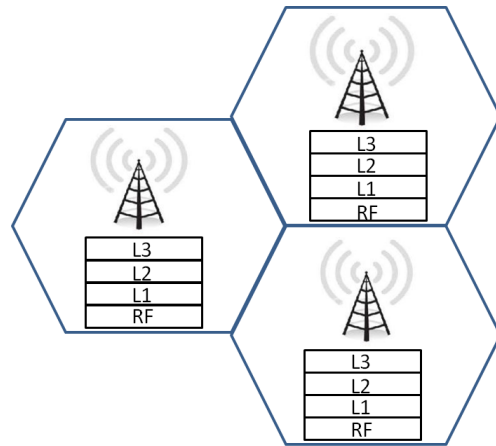


Fig. 1. Distributed architecture: Basedband and RRU co-located at cell site

## II. IMPACT OF FRONTHAUL LATENCY ON DL PERFORMANCE

In the downlink, without fronthaul delay, the earliest a HARQ process [5], [6] is available for a fresh or a re-transmission is 8 ms from the previous transmission on that HARQ process, after taking into account the processing times at the eNB and the UE, the over-the-air transmission times for downlink packets and uplink ack/nacks. LTE allows only

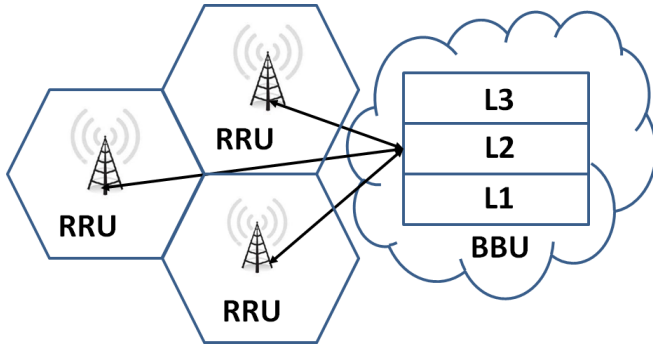


Fig. 2. Centralized Architecture: Pooled baseband at remote location

a maximum of eight HARQ processes, which implies that there can be at most eight outstanding packets without HARQ ack/nack feedback. Figure 3 shows that, for a cell with only one active user and full-buffer transmission, there will be no transmission on sub-frames where the user becomes ineligible for scheduling because of unavailable HARQ processes or HARQ stall. A round-trip delay of  $D$  ms implies that the HARQ ack/nack is received  $D/2$  ms later at the BBU and the scheduler would have to run  $D/2$  ms earlier. Therefore, for a single active UE, a two-way fronthaul latency of  $D$  ms implies that the UE cannot get scheduled for a duration of  $D$  ms over a window of length  $D + 8$  ms and there will be a degradation in single-UE peak throughput performance by a factor of  $D/(D+8)$ . However, when there are a larger number of active UEs, other UEs can be scheduled when a given UE is ineligible due to non-availability of HARQ processes. In addition to the HARQ stall issue, system performance is degraded due to use of outdated CSI [5], [7], [8]. The CSI information is outdated on two counts - one because it is available at the BBU  $D/2$  ms later than when it was reported at the RRU, and secondly, for a given TTI, the BBU needs to schedule  $D/2$  ms earlier than the distributed RAN case to account for the fronthaul delay. Our simulation results show that the performance degradation due to delayed CSI information is much less when compared to the performance degradation because of users becoming ineligible for scheduling.

To study the degradation in performance as a function of the number of active UEs and the fronthaul delay, we have conducted realistic multi-cell system simulations. Our simulations are compliant with 3GPP Release 11 specifications. We use proportionally fair scheduling scheme and frequency selective RB allocation using UE-reported sub-band CSI. Other simulation parameters and settings are given in Table I.

In Figure 4 we show the percentage loss in the geometric mean DL throughputs for both full-buffer and bursty traffic. The geometric mean of UE throughputs can be used as a single metric to compare the performance of the system for the PF scheduler, because the PF scheduler's objective is maximization of the geometric mean of UE throughputs, as described in [11]. The loss in performance due to non-

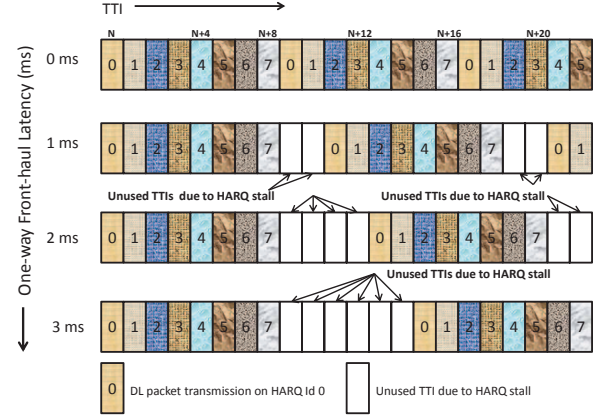


Fig. 3. HARQ stall due to non-ideal fronthaul

TABLE I  
SIMULATION PARAMETERS

|                                               |                                                                                          |
|-----------------------------------------------|------------------------------------------------------------------------------------------|
| Bandwidth                                     | 10 MHz                                                                                   |
| Carrier Frequency                             | 2 GHz                                                                                    |
| Propagation scenario                          | 3GPP Macro Case 1, 500 m ISD                                                             |
| Cell Topology                                 | 21 Cells (7 sites with 3 sectors each) with wrap around                                  |
| Pathloss model                                | $128.1 + 37.6 \log(R)$ , $R$ is distance in km                                           |
| Shadow fading                                 | Log normal with mean 0 and standard deviation of 8 dB                                    |
| Channel profile                               | SCM UMa (8 deg. azimuth spread)                                                          |
| Antenna Configuration                         | 2x2, single pair of cross-pol antennas both at Tx and Rx                                 |
| Macro Transmit Power                          | 40W (46 dBm)                                                                             |
| UE speed                                      | 3 km/h                                                                                   |
| UE Noise Figure                               | 9 dB                                                                                     |
| Traffic                                       | Full buffer (FB), bursty traffic                                                         |
| Scheduler                                     | Single-User MIMO, Frequency-selective scheduling (RB and MCS allocation)                 |
| Number of UEs per sector                      | Random UE Drop: Average of 5, 10, or 15 UEs per cell, exact number in each cell may vary |
| CSI feedback and period                       | PUSCH Mode 3-1, sub-band CSI feedback every 40 ms                                        |
| HO margin                                     | 2 dB                                                                                     |
| CoMP cluster                                  | Liquid cluster [10]                                                                      |
| CoMP set                                      | Best 3 cells within the CoMP cluster                                                     |
| CoMP coordination latency for centralized RAN | 0 ms                                                                                     |
| CoMP coordination latency for distributed RAN | 0-5 ms                                                                                   |
| PDCCH                                         | Ideal                                                                                    |
| Channel estimation                            | Ideal knowledge with no delay                                                            |
| RE overhead                                   | 4 out of 14 symbols                                                                      |
| Target Block Error Rate                       | 10 %                                                                                     |

ideal fronthaul delay is not only due to wasted resources because of HARQ stall and use of outdated CSI information for scheduling, but also due to the loss in multi-user diversity (MUD) gains because of fewer number of users eligible for scheduling per TTI. As expected, we see that increased fronthaul latency results in poorer performance for both full-buffer and bursty traffic models. We observe that the degradation in performance becomes smaller as the number of active UEs increase. With 5 or more active UEs and full-buffer traffic the loss in performance for a one-way fronthaul delay of 5 ms is 6% or less, compared to almost a 60% loss in geometric mean throughput for the single-UE case. We note that the degradation for bursty traffic is more than that of full-buffer traffic because with bursty traffic the number of eligible users in a sub-frame is significantly less than the number of users in a cell. Again, the degradation reduces with increase in offered load from 7 Mbps to 8 Mbps as the number of eligible users in a sub-frame is more for higher offered loads. Our results show that even with reasonably small number of active UEs, the loss in performance becomes marginal.

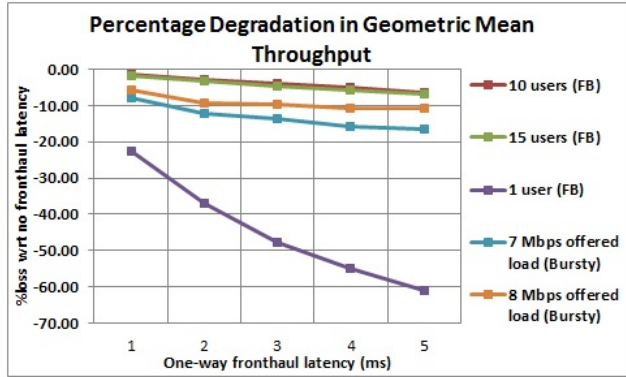


Fig. 4. Percentage loss in DL geometric mean UE throughputs for full-buffer (FB) and bursty traffic as a function of number of active UEs and fronthaul latency

With baseband pooling there is a near-zero coordination latency between different cells for CoMP techniques, such as, coordinated scheduling. However, with non-ideal fronthaul between BBU and RRU this comes at the cost of HARQ stall due to the longer round-trip-time between the UE and eNB HARQ entities. We have analyzed the performance of Multi-cell Coordinated Scheduling (MuCCS), a CoMP scheme described in [10] in the presence of fronthaul delay and as a function of number of active UEs. In Figure 5, we plot the percentage gains in the geometric mean of DL UE throughputs with full-buffer traffic over the corresponding no coordination throughput values. When computing the benefit and penalty values for the MuCCS scheme [10], only the UEs with available HARQ processes are taken into account. We find that there is a small reduction in MuCCS gains with increasing fronthaul latency, especially with large number of active users. We observe that, for an average of 15 users/cell, MuCCS gains are nearly the same for both 1 ms and 5 ms

one-way fronthaul latencies. However, for simulations with an average of 5 users/cell, there is a 2% drop in MuCCS gains for 5 ms one-way fronthaul latency when compared to 1 ms fronthaul latency. This is because muting gains are smaller when there are fewer active UEs in neighbor cells. In contrast, with larger number of active UEs, there is larger load-imbalance across cells which can be better exploited by coordinated scheduling.

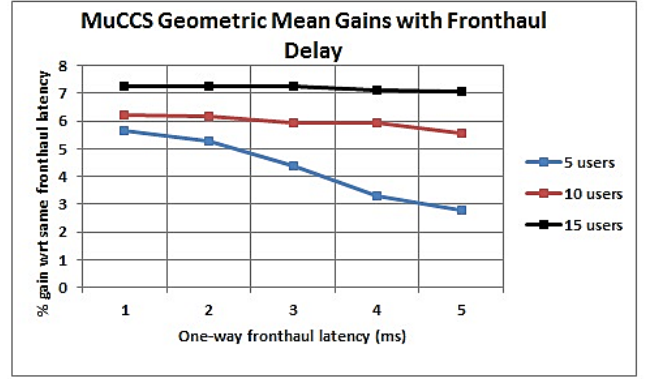


Fig. 5. Percentage gain due to MuCCS in the geometric mean of DL UE throughputs over the corresponding no coordination throughputs as the baseline

### III. DL PERFORMANCE FOR DISTRIBUTED AND CENTRALIZED RAN WITH DELAY

In this section, we compare the DL MuCCS performance in the presence of non-ideal fronthaul for centralized architecture and non-ideal backhaul for a distributed architecture. Figure 6 illustrates distributed and centralized coordination, where the former incurs backhaul delay in coordination between different baseband units, and the latter incurs fronthaul delay between the L1 and RF unit. A non-zero fronthaul delay causes HARQ stall, and use of outdated CSI information at the BBU further impacts scheduling and MCS selection for a user. In comparison, for a distributed architecture there is no HARQ stall because there is no additional delay for the HARQ feedback or in scheduling. However, the non-ideal interconnect between inter-site neighbors of a cluster results in a non-zero delay in exchange of benefit and penalty metrics used for multi-cell coordination. In contrast, the most recent benefit and penalty metrics can be used for intra-site neighbors as they have co-located schedulers. Therefore, for distributed RAN, the performance impact is because of using outdated metrics for *inter-site* coordination where as for centralized RAN, there is no coordination latency but there is degradation in performance because of HARQ stall.

In Figure 7, we compare the gains in geometric mean of UE throughputs for MuCCS with full-buffer traffic and centralized RAN versus that of distributed RAN for an average of 5 and 15 active UEs per cell in presence of non-ideal fronthaul and backhaul, respectively. The x-axis in Figure 7 is the one-way fronthaul delay, which is also assumed to be equal to the one-way backhaul delay, and the y-axis plots the

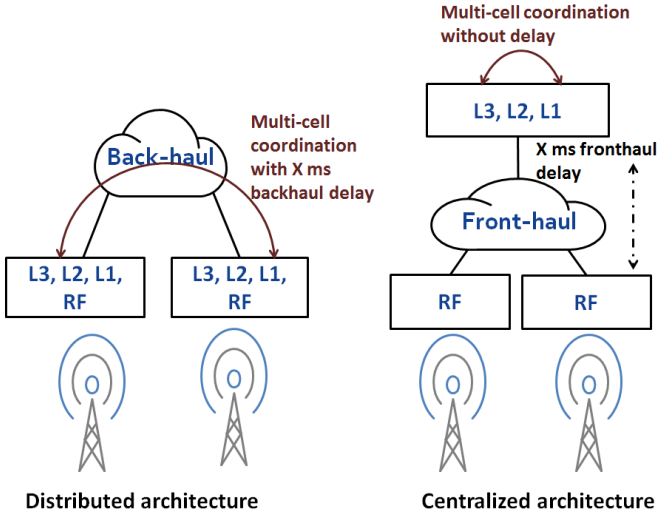


Fig. 6. Centralized (incurring fronthaul latency) vs. Distributed (incurring backhaul latency) coordination

gains due to coordinated scheduling over the baseline of no coordination and no fronthaul or backhaul delay. From these results a reduction in MuCCS gains is observed with delay for both centralized and distributed RAN. We observe that for all considered loads and delay values, distributed RAN performs better than centralized RAN with a larger difference in performance for larger delay values. Our simulation results for an average of 10 users/cell also show similar trends. However, they have been omitted here because of space constraints. The reason distributed RAN performs better than centralized RAN is because it can fully mimic the behavior of centralized RAN for one round of coordination message exchanges by not using the latest ack/nack and CQI information that arrives between the time of computation of benefit metric and the time of scheduling. In distributed RAN each scheduler computes the benefit metric one round-trip-time ahead of time so that the coordination decision from its neighbor is known at the time of the actual scheduling. Normally, the distributed RAN can use the latest ack/nack and CQI information at the time of scheduling. However, we can choose not to use this information to emulate the behaviour of centralized RAN which does not have access to this information. Thus, without using these advantages, the distributed RAN can be made to fully mimic the behavior of centralized RAN. However, if we do use these advantages of the distributed RAN, we expect distributed RAN to out-perform centralized RAN, which is what is observed in Figure 7.

#### IV. IMPACT OF FRONTHAUL LATENCY ON UL PERFORMANCE

The impact of fronthaul latency on UL performance is different from that in DL because UL employs synchronous retransmissions. Synchronous retransmissions for a packet whose fresh transmission occurs in TTI  $n$  can only occur in TTIs  $n+8$ ,  $n+16$ ,  $n+24$ , etc. Any non-zero fronthaul latency

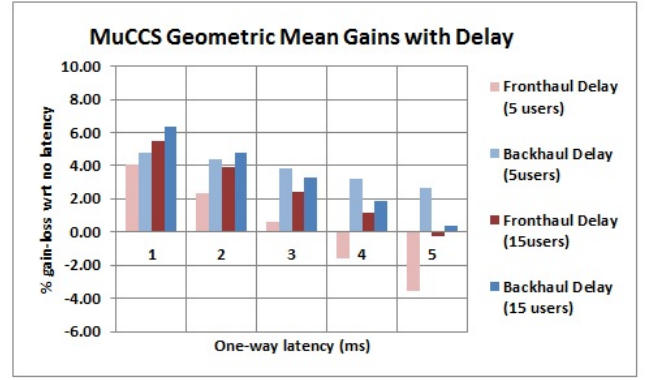


Fig. 7. Comparison of percentage gain-loss in the geometric mean of DL UE throughputs with fronthaul and backhaul latency for an average of 5, 15 users/cell

results in non-availability of the same HARQ process ID for transmission/retransmission 8 ms later. This leads to a 50% single UE peak throughput degradation. Figure 8 compares the degradation in single user throughput for both downlink and uplink for one-way delay upto 5 ms. From Figure 8 we observe that unlike in DL, there is a 50% loss in single UE peak throughput for all non-zero values of fronthaul round-trip latency up to 8 ms. Hence the impact is more severe, especially in low-load UL scenarios. For fronthaul latencies beyond 8 ms but less than 16 ms, the performance drops to 33% of the performance without fronthaul latency. However, the figures for one-way latency larger than 5 ms have been omitted due to space constraints. In general, for a fronthaul round-trip delay between  $8 \cdot n$  ms and  $8 \cdot (n+1)$  ms, the performance drops to  $100/(n+2)\%$  of the performance without fronthaul latency.

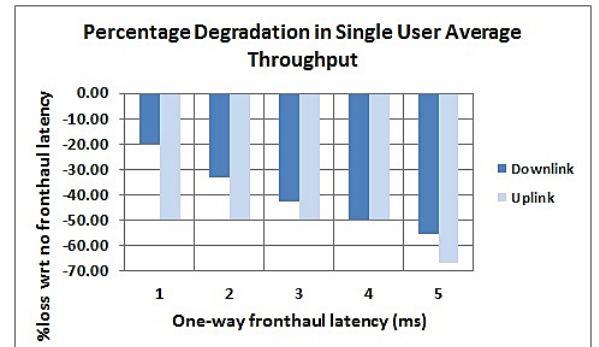


Fig. 8. Percentage loss in average UE throughput with fronthaul latency for downlink and uplink

Similar to DL, some of this performance loss is mitigated when there are larger number of active UEs. However, unlike in DL, additional enhancements are needed in UL to overcome the degradation because of synchronous retransmissions and power limitation of cell-edge UEs that prevent them from using all the RBs. We propose an enhanced *look-ahead scheduling* scheme that limits the number of scheduled UEs per subframe. This will allow some UEs to preserve their



HARQ processes for future TTIs so that larger resource utilization can be achieved. The *look-ahead scheduling* scheme works as follows: If the number of UEs with data to be transmitted is less than a threshold, then the number of scheduled UEs per subframe is limited to half the number of UEs with data to transmit, assuming the fronthaul latency is non-zero but less than or equal to 8 ms. If the fronthaul delay is between 8 and 16 ms, then the number of scheduled UEs is limited to one-third of the number of UEs with data to transmit. From Figure 9 we observe that, as seen in DL, the performance degradation reduces with increase in number of active users. This is because, with increase in number of active users, when a user is ineligible for scheduling due to a HARQ stall, there are other users available for scheduling. This plot is valid for all non-zero fronthaul round-trip latencies of 8 ms or less. The threshold for number of users scheduled in the enhanced scheme is chosen large enough that all cells schedule at most half its UEs in a given subframe. We find that the enhanced scheme results in significant improvement in performance. The reason for this performance improvement is that, with the enhanced scheme, the RB utilization increases significantly especially for smaller number of active UEs as seen in Figure 10.

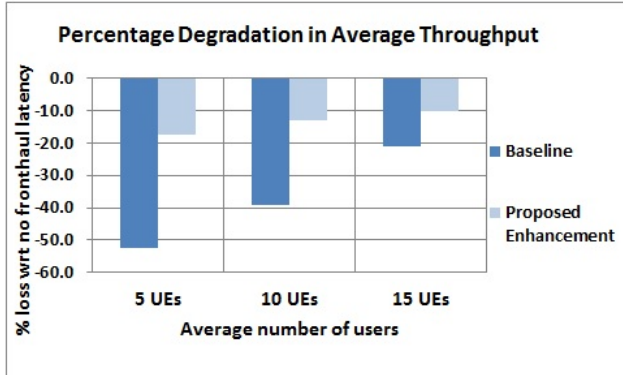


Fig. 9. Percentage loss in the UL average UE throughputs as a function of number of active UEs for the baseline and enhanced schemes

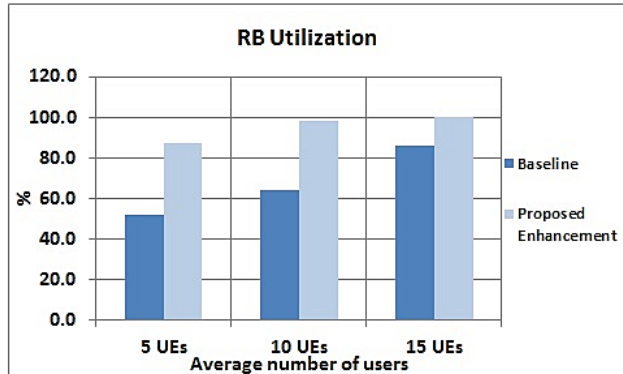


Fig. 10. UL RB utilization as a function of number of active UEs for the baseline and enhanced schemes

## V. CONCLUSIONS

In this paper, we have studied the impact of non-ideal fronthaul on the downlink and uplink performance of LTE-Advanced networks. We showed that, while the impact on the peak throughput is quite severe, it is more modest when we have a larger number of active UEs. We showed that the degradation in performance is larger in the uplink than in the downlink due to the synchronous HARQ mechanism used in LTE uplink. In the downlink, we compared the performance gains with coordinated scheduling for both distributed and centralized RAN as a function of backhaul and fronthaul delay, respectively and with different number of active users in the system. Our results show that for all considered loads and delay values distributed RAN performs better than centralized RAN. In the uplink, we developed a novel *look-ahead scheduling* scheme that reduces the number of scheduled UEs per subframe to minimize the loss in performance due to HARQ stall. We showed that the enhanced scheme results in significant improvement in performance, especially when the number of active UEs is small.

## VI. ACKNOWLEDGMENT

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